



МИНОБРНАУКИ РОССИИ

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РТУ МИРЭА

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Кафедра прикладной математики (ПМ)**

ОТЧЕТ ПО ПРАКТИЧЕСКОЙ РАБОТЕ №2
по дисциплине «Системы управления большими данными»

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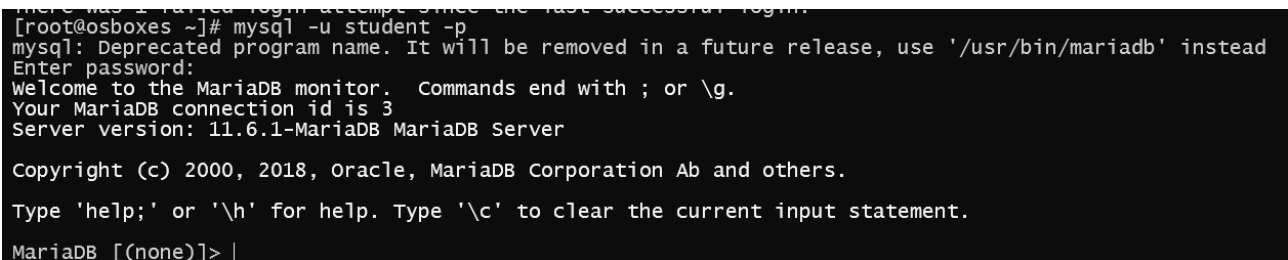
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1 СОЗДАНИЕ БД И НАПОЛНЕНИЕ ДАННЫМИ

1.1 Подключение к MariaDB

Процесс подключения к MariaDB представлен на Рисунке 1.



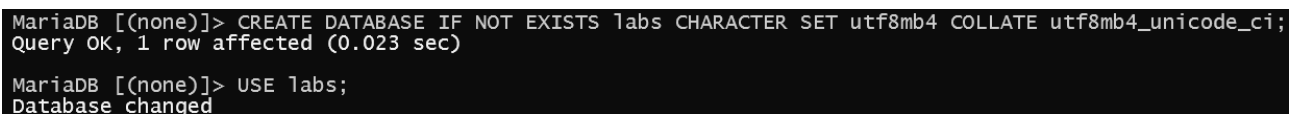
```
There was 1 failed login attempt since the last successful login.  
[root@osboxes ~]# mysql -u student -p  
mysql: Deprecated program name. It will be removed in a future release, use '/usr/bin/mariadb' instead  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 3  
Server version: 11.6.1-MariaDB MariaDB Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
MariaDB [(none)]> |
```

Рисунок 1 — Подключение к MariaDB

Перейдем к созданию БД.

1.2 Создание БД

Создание базы данных представлено на Рисунке 2.



```
MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS labs CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;  
Query OK, 1 row affected (0.023 sec)  
  
MariaDB [(none)]> USE labs;  
Database changed
```

Рисунок 2 — Создание базы данных

Создание таблиц в БД представлено на Рисунке 3.

```

MariaDB [labs]> DROP TABLE IF EXISTS authors;
Query OK, 0 rows affected, 1 warning (0.055 sec)

MariaDB [labs]> CREATE TABLE authors (
  id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(120) UNIQUE,
  birthdate DATE,
  added DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  INDEX idx_name (last_name, first_name)
) ENGINE=InnoDB;
Query OK, 0 rows affected (0.057 sec)

MariaDB [labs]> DROP TABLE IF EXISTS posts;
Query OK, 0 rows affected, 1 warning (0.009 sec)

MariaDB [labs]> CREATE TABLE posts (
  id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
  author_id INT NOT NULL,
  title VARCHAR(200),
  content TEXT,
  published_at DATETIME,
  CONSTRAINT fk_posts_author FOREIGN KEY (author_id) REFERENCES authors(id)
  ON DELETE CASCADE ON UPDATE CASCADE,
  INDEX idx_published (published_at)
) ENGINE=InnoDB;
Query OK, 0 rows affected (0.020 sec)

MariaDB [labs]> |

```

Рисунок 3 — Создание таблиц

Заполнение таблиц данными представлено на Рисунке 4.

```

MariaDB [labs]> INSERT INTO authors (first_name, last_name, email, birthdate, added) VALUES
-> ('Dorthy', 'Hicks', 'dorthy.hicks@example.com', '1990-02-14', '2023-05-10 09:15:00'),
-> ('Ivan', 'Petrov', 'ivan.petrov@example.com', '1987-11-02', '2023-05-11 10:20:00'),
-> ('Anna', 'Smirnova', 'anna.smirnova@example.com', '1995-07-30', '2023-05-12 11:25:00'),
-> ('John', 'Doe', 'john.doe@example.com', '1980-01-05', '2023-05-13 12:30:00'),
-> ('Maria', 'Garcia', 'maria.garcia@example.com', '1992-03-18', '2023-05-14 13:35:00'),
-> ('Wei', 'Zhang', 'wei.zhang@example.com', '1993-09-09', '2023-05-15 14:40:00'),
-> ('Fatima', 'Ali', 'fatima.ali@example.com', '1988-12-21', '2023-05-16 15:45:00'),
-> ('Elena', 'Ivanova', 'elena.ivanova@example.com', '1996-06-11', '2023-05-17 16:50:00');
Query OK, 8 rows affected (0.022 sec)
Records: 8 Duplicates: 0 Warnings: 0

MariaDB [labs]> INSERT INTO posts (author_id, title, content, published_at) VALUES
-> (1, 'Hello Sqoop', 'Intro to Sqoop import/export', '2023-06-01 09:00:00'),
-> (2, 'HDFS Basics', 'Blocks, replication, CLI', '2023-06-02 10:00:00'),
-> (3, NULL, 'Untitled draft content', '2023-06-03 11:00:00'),
-> (4, 'Parquet FTW', 'Why Parquet is efficient', '2023-06-04 12:00:00'),
-> (5, 'Compression', 'Gzip vs Snappy vs LZ0', '2023-06-05 13:00:00'),
-> (6, NULL, 'Another draft without title', '2023-06-06 14:00:00'),
-> (7, 'Filters', 'Using WHERE in Sqoop', '2023-06-07 15:00:00'),
-> (8, 'Columns', 'Import selected columns only', '2023-06-08 16:00:00');
Query OK, 8 rows affected (0.004 sec)
Records: 8 Duplicates: 0 Warnings: 0

MariaDB [labs]> |

```

Рисунок 4 — Заполнение таблиц

Проверка корректности создания представлена на Рисунке 5.

```
MariaDB [labs]> SELECT COUNT(*) AS authors_cnt FROM authors;
+-----+
| authors_cnt |
+-----+
|          8 |
+-----+
1 row in set (0.005 sec)

MariaDB [labs]> SELECT COUNT(*) AS posts_cnt FROM posts;
+-----+
| posts_cnt |
+-----+
|          8 |
+-----+
1 row in set (0.002 sec)

MariaDB [labs]> |
```

Рисунок 5 — Проверка создания

Таким образом, база данных создана и заполнена.

1.3 Быстрый обзор данных

Обзор данных представлен на Рисунке 6.

```

MariaDB [labs]> SHOW TABLES;
+-----+
| Tables_in_labs |
+-----+
| authors        |
| posts          |
+-----+
2 rows in set (0.001 sec)

MariaDB [labs]> DESC authors;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default          | Extra          |
+-----+-----+-----+-----+-----+-----+
| id         | int(11)       | NO   | PRI | NULL             | auto_increment |
| first_name | varchar(50)   | NO   |     | NULL             |                |
| last_name  | varchar(50)   | NO   |     | NULL             |                |
| email      | varchar(120)  | YES  |     | NULL             |                |
| birthdate  | date          | YES  |     | NULL             |                |
| added      | datetime      | NO   |     | current_timestamp() |                |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.007 sec)

MariaDB [labs]> DESC posts;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default          | Extra          |
+-----+-----+-----+-----+-----+-----+
| id         | int(11)       | NO   | PRI | NULL             | auto_increment |
| author_id  | int(11)       | NO   | MUL | NULL             |                |
| title      | varchar(200)  | YES  |     | NULL             |                |
| content    | text          | YES  |     | NULL             |                |
| published_at | datetime      | YES  | MUL | NULL             |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.002 sec)

MariaDB [labs]> SELECT id, first_name, last_name, email, added FROM authors LIMIT 5;
+-----+-----+-----+-----+-----+
| id | first_name | last_name | email                      | added          |
+-----+-----+-----+-----+-----+
| 1  | Dorothy   | Hicks    | dorthy.hicks@example.com  | 2023-05-10 09:15:00 |
| 2  | Ivan      | Petrov   | ivan.petrov@example.com   | 2023-05-11 10:20:00 |
| 3  | Anna      | Smirnova | anna.smirnova@example.com  | 2023-05-12 11:25:00 |
| 4  | John      | Doe      | john.doe@example.com       | 2023-05-13 12:30:00 |
| 5  | Maria     | Garcia   | maria.garcia@example.com   | 2023-05-14 13:35:00 |
+-----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [labs]> SELECT id, author_id, title, published_at FROM posts ORDER BY id LIMIT 5;
+-----+-----+-----+-----+
| id | author_id | title          | published_at          |
+-----+-----+-----+-----+
| 1  | 1         | Hello Sqoop   | 2023-06-01 09:00:00  |
| 2  | 2         | HDFS Basics  | 2023-06-02 10:00:00  |
| 3  | 3         | NULL          | 2023-06-03 11:00:00  |
| 4  | 4         | Parquet FTW   | 2023-06-04 12:00:00  |
| 5  | 5         | Compression   | 2023-06-05 13:00:00  |
+-----+-----+-----+-----+
5 rows in set (0.002 sec)

MariaDB [labs]> |

```

Рисунок 6 — Обзор данных

Выйдем из MariaDB командой quit.

2 БАЗОВЫЕ КОМАНДЫ SQOOP И ПРОСМОТР МЕТАДААННЫХ

Вызов команды sqoop help представлен на Рисунке 7.

```
[root@osboxes ~]# sqoop help
Warning: /usr/local/sqoop/sqoop-1.4.7/./hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2025-09-30 13:30:20,505 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
usage: sqoop COMMAND [ARGS]

Available commands:
codegen          Generate code to interact with database records
create-hive-table Import a table definition into Hive
eval            Evaluate a SQL statement and display the results
export          Export an HDFS directory to a database table
help            List available commands
import          Import a table from a database to HDFS
import-all-tables Import tables from a database to HDFS
import-mainframe Import datasets from a mainframe server to HDFS
job             Work with saved jobs
list-databases  List available databases on a server
list-tables    List available tables in a database
merge          Merge results of incremental imports
metastore       Run a standalone Sqoop metastore
version         Display version information

See 'sqoop help COMMAND' for information on a specific command.
[root@osboxes ~]# sqoop help import
Warning: /usr/local/sqoop/sqoop-1.4.7/./hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2025-09-30 13:30:28,890 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
usage: sqoop import [GENERIC-ARGS] [TOOL-ARGS]

Common arguments:
--connect <jdbc-uri>          Specify JDBC
                             connect
                             string
--connection-manager <class-name> Specify
                             connection
                             manager
                             class name
--connection-param-file <properties-file> Specify
                             connection
                             parameters
                             file
--driver <class-name>        Manually
                             specify JDBC
                             driver class
                             to use
--hadoop-home <hdir>         Override
```

Рисунок 7 — Команды sqoop

Список баз данных при подключении с помощью sqoop представлен на Рисунке 8.

```
[root@osboxes ~]# sqoop list-databases --connect jdbc:mysql://localhost --username student -P
Warning: /usr/local/sqoop/sqoop-1.4.7/./hcatalog does not exist! Hcatalog jobs will fail.
Please set $HCAT_HOME to the root of your Hcatalog installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2025-09-30 13:31:41,546 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
Enter password:
2025-09-30 13:31:44,346 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
Tue Sep 30 13:31:44 EDT 2025 WARN: Establishing SSL connection without server's identity verification is not recommended. According to MySQL 5.5.45+, 5.6.26
+ and 5.7.6+ requirements SSL connection must be established by default if explicit option isn't set. For compliance with existing applications not using SS
L the verifyServerCertificate property is set to 'false'. You need either to explicitly disable SSL by setting useSSL=false, or set useSSL=true and provide
truststore for server certificate verification.
information_schema
mysql
performance_schema
sys
labs
Tue Sep 30 13:31:45 EDT 2025 WARN: Caught while disconnecting...
```

Рисунок 8 — Список баз данных

Таблицы в БД “labs” представлены на Рисунке 9.

```
[root@osboxes ~]# sqoop list-tables --connect jdbc:mysql://localhost/labs --username student -P
Warning: /usr/local/sqoop/sqoop-1.4.7/./hcatalog does not exist! Hcatalog jobs will fail.
Please set $HCAT_HOME to the root of your Hcatalog installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /usr/local/sqoop/sqoop-1.4.7/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2025-09-30 13:34:12,410 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
Enter password:
2025-09-30 13:34:16,091 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
Tue Sep 30 13:34:16 EDT 2025 WARN: Establishing SSL connection without server's identity verification is not recommended. According to MySQL 5.5.45+, 5.6.26
+ and 5.7.6+ requirements SSL connection must be established by default if explicit option isn't set. For compliance with existing applications not using SS
L the verifyServerCertificate property is set to 'false'. You need either to explicitly disable SSL by setting useSSL=false, or set useSSL=true and provide
truststore for server certificate verification.
authors
posts
Tue Sep 30 13:34:16 EDT 2025 WARN: Caught while disconnecting...
```

Рисунок 9 — Таблицы в labs

Таким образом, можно подключаться к MariaDB с помощью sqoop.

3 ИМПОРТ В HDFS (ОСНОВЫ)

3.1 Импорт целой таблицы posts в домашний каталог HDFS

Процесс импорта таблицы posts представлен на Рисунке 10.

```
[root@osboxes lib]# sqoop import --driver org.mariadb.jdbc.Driver \
> --connect "jdbc:mariadb://localhost:3306/labs?useSSL=false&allowPublicKeyRetrieval=true&useUnicode=true&characterEncoding=UTF-8" \
> --username student -P --table posts --delete-target-dir
2025-09-30 14:30:11,498 INFO Sqoop.Sqoop: Running Sqoop version: 1.4.7
Enter password:
2025-09-30 14:30:14,232 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriate connection manager is not being set (via -
fy explicitly which connection manager should be used next time.
2025-09-30 14:30:14,245 INFO manager.SqlManager: Using default fetchSize of 1000
2025-09-30 14:30:14,245 INFO tool.CodeGenTool: Beginning code generation
2025-09-30 14:30:14,479 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM posts AS t WHERE 1=0
2025-09-30 14:30:14,493 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM posts AS t WHERE 1=0
2025-09-30 14:30:14,515 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/hadoop/hadoop
Note: /tmp/sqoop-root/compile/7c7084733dd3176a1fe45510a22c8d47/posts.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2025-09-30 14:30:15,836 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/7c7084733dd3176a1fe45510a22c8d47/posts.jar
2025-09-30 14:30:16,805 INFO tool.ImportTool: Destination directory posts deleted.
2025-09-30 14:30:16,821 INFO mapreduce.ImportJobBase: Beginning import of posts
2025-09-30 14:30:16,821 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2025-09-30 14:30:16,828 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2025-09-30 14:30:16,830 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM posts AS t WHERE 1=0
2025-09-30 14:30:16,840 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0001
2025-09-30 14:30:16,931 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2025-09-30 14:30:17,269 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1759256979608_0001
2025-09-30 14:30:20,380 INFO db.DBInputFormat: Using read committed transaction isolation
2025-09-30 14:30:20,381 INFO db.DataDrivenDBInputFormat: BoundingValsQuery: SELECT MIN(id), MAX(id) FROM posts
2025-09-30 14:30:20,385 INFO db.IntegerSplitter: Split size: 1; Num splits: 4 from: 1 to: 8
2025-09-30 14:30:20,461 INFO mapreduce.JobSubmitter: number of splits:4
2025-09-30 14:30:20,607 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0001
2025-09-30 14:30:20,607 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-09-30 14:30:20,909 INFO conf.Configuration: resource-types.xml not found
2025-09-30 14:30:20,910 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-09-30 14:30:21,304 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0001
2025-09-30 14:30:21,386 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0001/
2025-09-30 14:30:21,388 INFO mapreduce.Job: Running job: job_1759256979608_0001
2025-09-30 14:30:30,596 INFO mapreduce.Job: Job job_1759256979608_0001 running in uber mode : false
2025-09-30 14:30:30,598 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 14:30:38,990 INFO mapreduce.Job: map 25% reduce 0%
2025-09-30 14:30:40,000 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 14:30:41,027 INFO mapreduce.Job: Job job_1759256979608_0001 completed successfully
2025-09-30 14:30:41,125 INFO mapreduce.Job: Counters: 33
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1128292
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=377
  HDFS: Number of bytes written=484
  HDFS: Number of read operations=24
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=8
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=4
  Other local map tasks=4
  Total time spent by all maps in occupied slots (ms)=19644
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=19644
  Total vcore-milliseconds taken by all map tasks=19644
  Total megabyte-milliseconds taken by all map tasks=20115456
Map-Reduce Framework
  Map input records=8
  Map output records=8
  Input split bytes=377
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=419
  CPU time spent (ms)=4300
  Physical memory (bytes) snapshot=893730816
  Virtual memory (bytes) snapshot=11248979968
  Total committed heap usage (bytes)=745013248
  Peak Map Physical memory (bytes)=226988032
  Peak Map Virtual memory (bytes)=2814132224
File Input Format Counters
  Bytes Read=0
File Output Format Counters
```

Рисунок 10 — Импорт таблицы posts

Проверка корректного импорта представлена на Рисунке 11.

```

[root@osboxes lib]# hdfs dfs -ls /user/root/posts
Found 5 items
-rw-r--r--  1 root supergroup          0 2025-09-30 14:30 /user/root/posts/_SUCCESS
-rw-r--r--  1 root supergroup       130 2025-09-30 14:30 /user/root/posts/part-m-00000
-rw-r--r--  1 root supergroup       117 2025-09-30 14:30 /user/root/posts/part-m-00001
-rw-r--r--  1 root supergroup       119 2025-09-30 14:30 /user/root/posts/part-m-00002
-rw-r--r--  1 root supergroup       118 2025-09-30 14:30 /user/root/posts/part-m-00003
[root@osboxes lib]# hdfs dfs -cat /user/root/posts/part-m-00000
1,1,Hello Sqoop,Intro to Sqoop import/export,2023-06-01 09:00:00.0
2,2,HDFS Basics,Blocks, replication, CLI,2023-06-02 10:00:00.0
[root@osboxes lib]# hdfs dfs -cat /user/root/posts/part-m-00001
3,3,null,Untitled draft content,2023-06-03 11:00:00.0
4,4,Parquet FTW,Why Parquet is efficient,2023-06-04 12:00:00.0
[root@osboxes lib]# hdfs dfs -cat /user/root/posts/part-m-00002
5,5,Compression,Gzip vs Snappy vs LZ0,2023-06-05 13:00:00.0
6,6,null,Another draft without title,2023-06-06 14:00:00.0
[root@osboxes lib]# hdfs dfs -cat /user/root/posts/part-m-00003
7,7,Filters,Using WHERE in Sqoop,2023-06-07 15:00:00.0
8,8,Columns,Import selected columns only,2023-06-08 16:00:00.0
[root@osboxes lib]# |

```

Рисунок 11 — Проверка импорта

Перейдем к импорту authors

3.2 Импорт таблицы authors

Процесс импорта таблицы authors представлен на Рисунке 12.

```

2025-09-30 14:39:13,389 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriate connection manager is not being set (via --
fy explicitly which connection manager should be used next time.
2025-09-30 14:39:13,404 INFO manager.SqlManager: Using default fetchSize of 1000
2025-09-30 14:39:13,404 INFO tool.CodeGenTool: Beginning code generation
2025-09-30 14:39:13,625 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 14:39:13,638 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 14:39:13,660 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/hadoop/hadoop
Note: /tmp/sqoop-root/compile/57493e0c2c6c4ddd056d5812f8f8fecc/authors.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2025-09-30 14:39:14,937 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/57493e0c2c6c4ddd056d5812f8f8fecc/authors.jar
2025-09-30 14:39:14,964 INFO mapreduce.ImportJobBase: Beginning import of authors
2025-09-30 14:39:14,965 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2025-09-30 14:39:15,184 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2025-09-30 14:39:15,187 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 14:39:15,649 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2025-09-30 14:39:15,743 INFO client.DefaultHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2025-09-30 14:39:16,126 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1759256979608_0002
2025-09-30 14:39:18,945 INFO db.DBInputFormat: Using read committed transaction isolation
2025-09-30 14:39:18,946 INFO db.DataDrivenDBInputFormat: BoundingValsQuery: SELECT MIN(id), MAX(id) FROM authors
2025-09-30 14:39:18,949 INFO db.IntegerSplitter: Split size: 1; Num splits: 4 from: 1 to: 8
2025-09-30 14:39:19,017 INFO mapreduce.JobSubmitter: number of splits:4
2025-09-30 14:39:19,153 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0002
2025-09-30 14:39:19,154 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-09-30 14:39:19,391 INFO configuration: resource-types.xml not found
2025-09-30 14:39:19,391 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-09-30 14:39:19,452 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0002
2025-09-30 14:39:19,494 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0002/
2025-09-30 14:39:19,495 INFO mapreduce.Job: Running job: job_1759256979608_0002
2025-09-30 14:39:26,601 INFO mapreduce.Job: Job job_1759256979608_0002 running in uber mode : false
2025-09-30 14:39:26,602 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 14:39:31,737 INFO mapreduce.Job: map 25% reduce 0%
2025-09-30 14:39:32,941 INFO mapreduce.Job: map 50% reduce 0%
2025-09-30 14:39:34,980 INFO mapreduce.Job: map 75% reduce 0%
2025-09-30 14:39:35,989 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 14:39:36,001 INFO mapreduce.Job: Job job_1759256979608_0002 completed successfully
2025-09-30 14:39:36,091 INFO mapreduce.Job: Counters: 33
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1128156
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=377
  HDFS: Number of bytes written=568
  HDFS: Number of read operations=24
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=8
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=4
  Other local map tasks=4
  Total time spent by all maps in occupied slots (ms)=15306
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=15306
  Total vcore-milliseconds taken by all map tasks=15306
  Total megabyte-milliseconds taken by all map tasks=15673344
Map-Reduce Framework
  Map input records=8
  Map output records=8
  Input split bytes=377
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=372
  CPU time spent (ms)=3610
  Physical memory (bytes) snapshot=870944768
  Virtual memory (bytes) snapshot=11252084736
  Total committed heap usage (bytes)=822083584
  Peak Map Physical memory (bytes)=224075776
  Peak Map Virtual memory (bytes)=2813882368
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=568
2025-09-30 14:39:36,099 INFO mapreduce.ImportJobBase: Transferred 568 bytes in 20.4369 seconds (27.7928 bytes/sec)
2025-09-30 14:39:36,105 INFO mapreduce.ImportJobBase: Retrieved 8 records.
[root@osboxes lib]#

```

Рисунок 12 — Импорт таблицы authors

Проверка импорта представлена на Рисунке 13.

```

[root@osboxes lib]# hdfs dfs -ls /mywarehouse/authors
Found 5 items
-rw-r--r-- 1 root supergroup 0 2025-09-30 14:39 /mywarehouse/authors/_SUCCESS
-rw-r--r-- 1 root supergroup 144 2025-09-30 14:39 /mywarehouse/authors/part-m-00000
-rw-r--r-- 1 root supergroup 140 2025-09-30 14:39 /mywarehouse/authors/part-m-00001
-rw-r--r-- 1 root supergroup 140 2025-09-30 14:39 /mywarehouse/authors/part-m-00002
-rw-r--r-- 1 root supergroup 144 2025-09-30 14:39 /mywarehouse/authors/part-m-00003
[root@osboxes lib]# hdfs dfs -cat /mywarehouse/authors/part-m-00000 | head
1,Dorthy,Hicks,dorthy.hicks@example.com,1990-02-14,2023-05-10 09:15:00.0
2,Ivan,Petrov,ivan.petrov@example.com,1987-11-02,2023-05-11 10:20:00.0
[root@osboxes lib]# hdfs dfs -cat /mywarehouse/authors/part-m-00001 | head
3,Anna,Smirnova,anna.smirnova@example.com,1995-07-30,2023-05-12 11:25:00.0
4,John,Doe,john.doe@example.com,1980-01-05,2023-05-13 12:30:00.0
[root@osboxes lib]# hdfs dfs -cat /mywarehouse/authors/part-m-00002 | head
5,Maria,Garcia,maria.garcia@example.com,1992-03-18,2023-05-14 13:35:00.0
6,Wei,Zhang,wei.zhang@example.com,1993-09-09,2023-05-15 14:40:00.0
^[[A[root@osboxes lib]# hdfs dfs -cat /mywarehouse/authors/part-m-00003 | head
7,Fatima,Ali,fatima.ali@example.com,1988-12-21,2023-05-16 15:45:00.0
8,Elena,Ivanova,elena.ivanova@example.com,1996-06-11,2023-05-17 16:50:00.0
[root@osboxes lib]#

```

Рисунок 13 — Проверка импорта таблицы authors

Таким образом, таблицы из MariaDB импортированы в hdfs.

4 ВАРИАНТЫ ИМПОРТА

4.1 Импорт выбранных столбцов

Импорт выбранных столбцов представлен на Рисунке 14.

```
2025-09-30 14:55:38,595 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-09-30 14:55:38,660 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0003
2025-09-30 14:55:38,697 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0003/
2025-09-30 14:55:38,697 INFO mapreduce.Job: Running job: job_1759256979608_0003
2025-09-30 14:55:45,829 INFO mapreduce.Job: Job job_1759256979608_0003 running in uber mode : false
2025-09-30 14:55:45,830 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 14:55:52,935 INFO mapreduce.Job: map 50% reduce 0%
2025-09-30 14:55:53,944 INFO mapreduce.Job: map 75% reduce 0%
2025-09-30 14:55:54,951 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 14:55:54,964 INFO mapreduce.Job: Job job_1759256979608_0003 completed successfully
2025-09-30 14:55:55,064 INFO mapreduce.Job: Counters: 34
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1128104
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=377
  HDFS: Number of bytes written=288
  HDFS: Number of read operations=24
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=8
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Killed map tasks=1
  Launched map tasks=4
  Other local map tasks=4
  Total time spent by all maps in occupied slots (ms)=17840
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=17840
  Total vcore-milliseconds taken by all map tasks=17840
  Total megabyte-milliseconds taken by all map tasks=18268160
Map-Reduce Framework
  Map input records=8
  Map output records=8
  Input split bytes=377
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=409
  CPU time spent (ms)=4190
  Physical memory (bytes) snapshot=917839872
  Virtual memory (bytes) snapshot=11265007616
  Total committed heap usage (bytes)=704118784
  Peak Map Physical memory (bytes)=245661696
  Peak Map Virtual memory (bytes)=2828165120
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=288
2025-09-30 14:55:55,072 INFO mapreduce.ImportJobBase: Transferred 288 bytes in 22.387 seconds (12.8646 bytes/sec)
2025-09-30 14:55:55,077 INFO mapreduce.ImportJobBase: Retrieved 8 records.
[root@osboxes lib]# hdfs dfs -cat /user/student/authors_cols/part-m-00000 | head
Dorothy Hicks dorothy.hicks@example.com
Ivan Petrov ivan.petrov@example.com
[root@osboxes lib]#
```

Рисунок 14 — Импорт выбранных столбцов

4.2 Импорт по условию (WHERE)

Импорт по условию представлен на Рисунке 15.

```

st_name='Dorthy'" --target-dir /user/student/authors_Dorthy
2025-09-30 15:30:10,788 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2025-09-30 15:30:10,837 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider u
2025-09-30 15:30:10,909 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriat
fy explicitly which connection manager should be used next time.
2025-09-30 15:30:10,918 INFO manager.SqlManager: Using default fetchSize of 1000
2025-09-30 15:30:10,918 INFO tool.CodeGenTool: Beginning code generation
2025-09-30 15:30:11,221 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 15:30:11,240 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 15:30:11,274 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/hadoop/hadoop
Note: /tmp/sqoop-root/compile/2d7312855d937962a2f870c50929fb0b/authors.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2025-09-30 15:30:13,382 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/2d7312855d937962a2f
2025-09-30 15:30:13,432 INFO mapreduce.ImportJobBase: Beginning import of authors
2025-09-30 15:30:13,434 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.j
2025-09-30 15:30:13,686 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2025-09-30 15:30:13,691 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 15:30:14,261 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job
2025-09-30 15:30:14,444 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:
2025-09-30 15:30:15,251 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/st
2025-09-30 15:30:19,501 INFO db.DBInputFormat: Using read committed transaction isolation
2025-09-30 15:30:19,502 INFO db.DataDrivenDBInputFormat: BoundingValsQuery: SELECT MIN(id), MAX(id) FROM authors W
2025-09-30 15:30:19,515 INFO db.IntegerSplitter: Split size: 0; Num splits: 4 from: 1 to: 1
2025-09-30 15:30:19,994 INFO mapreduce.JobSubmitter: number of splits:1
2025-09-30 15:30:20,168 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0004
2025-09-30 15:30:20,168 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-09-30 15:30:20,454 INFO conf.Configuration: resource-types.xml not found
2025-09-30 15:30:20,454 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-09-30 15:30:20,551 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0004
2025-09-30 15:30:20,621 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759
2025-09-30 15:30:20,624 INFO mapreduce.Job: Running job: job_1759256979608_0004
2025-09-30 15:30:29,865 INFO mapreduce.Job: Job job_1759256979608_0004 running in uber mode : false
2025-09-30 15:30:29,866 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 15:30:35,962 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 15:30:36,984 INFO mapreduce.Job: Job job_1759256979608_0004 completed successfully
2025-09-30 15:30:37,079 INFO mapreduce.Job: Counters: 33
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=282175
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=95
  HDFS: Number of bytes written=73
  HDFS: Number of read operations=6
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=1
  Other local map tasks=1
  Total time spent by all maps in occupied slots (ms)=3964
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=3964
  Total vcore-milliseconds taken by all map tasks=3964
  Total megabyte-milliseconds taken by all map tasks=4059136
Map-Reduce Framework
  Map input records=1
  Map output records=1
  Input split bytes=95
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=90
  CPU time spent (ms)=1170
  Physical memory (bytes) snapshot=225280000
  Virtual memory (bytes) snapshot=2810662912
  Total committed heap usage (bytes)=181403648
  Peak Map Physical memory (bytes)=225280000
  Peak Map Virtual memory (bytes)=2810662912
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=73
2025-09-30 15:30:37,088 INFO mapreduce.ImportJobBase: Transferred 73 bytes in 22.8094 seconds (3.2004 bytes/sec)
2025-09-30 15:30:37,094 INFO mapreduce.ImportJobBase: Retrieved 1 records.
[root@osboxes lib]# hdfs dfs -cat /user/student/authors_Dorthy/* | head
1 Dorthy Hicks dorthy.hicks@example.com 1990-02-14 2023-05-10 09:15:00.0
[root@osboxes lib]# |

```

Рисунок 15 — Импорт по условию

4.3 Импорт в формат Parquet

Импорт в формат Parquet представлен на Рисунке 16.

Spexrk

```
Using Python version 3.8.11 (default, Sep 2 2025 07:20:45)
Spark context Web UI available at http://10.192.3.221:4041
Spark context available as 'sc' (master = local[*], app id = local-1759262245785).
SparkSession available as 'spark'.
>>> df = spark.read.parquet("/mywarehouse/authors_parquet")
>>> df.printSchema()
root
  |-- id: integer (nullable = true)
  |-- first_name: string (nullable = true)
  |-- last_name: string (nullable = true)
  |-- email: string (nullable = true)
  |-- birthdate: long (nullable = true)
  |-- added: long (nullable = true)

>>> df.show()
+-----+-----+-----+-----+-----+-----+
| id|first_name|last_name|email| birthdate| added|
+-----+-----+-----+-----+-----+-----+
| 7|Fatima|Ali|fatima.ali@example...|598683600000|1684266300000|
| 8|Elena|Ivanova|elena.ivanova@exa...|834465600000|1684356600000|
| 1|Dorthy|Hicks|dorthy.hicks@exam...|634971600000|1683724500000|
| 2|Ivan|Petrov|ivan.petrov@examp...|562827600000|1683814800000|
| 5|Maria|Garcia|maria.garcia@exam...|700894800000|1684085700000|
| 6|Wei|Zhang|wei.zhang@example...|747547200000|1684176000000|
| 3|Anna|Smirnova|anna.smirnova@exa...|807076800000|1683905100000|
| 4|John|Doe|john.doe@example.com|315896400000|1683995400000|
+-----+-----+-----+-----+-----+-----+

>>>
```

4.4 Импорт со сжатием

15

```

[root@osboxes lib]# sqoop import --driver org.mariadb.jdbc.Driver --connect jdbc:mariadb://localhost/labs --username student --password studen
t --table authors --target-dir /mywarehouse/authors_compressed --compress
2025-09-30 16:00:56,662 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2025-09-30 16:00:56,703 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2025-09-30 16:00:56,773 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriate connection manager is not bein
g used next time
2025-09-30 16:00:56,787 INFO manager.SqlManager: Using default fetchSize of 1000
2025-09-30 16:00:56,787 INFO tool.CodeGenTool: Beginning code generation
2025-09-30 16:00:57,019 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 16:00:57,033 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 16:00:57,052 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/hadoop/hadoop
Note: /tmp/sqoop-root/compile/1967771a03f6dff0310803c8f8193d9e/authors.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2025-09-30 16:00:58,339 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/1967771a03f6dff0310803c8f8193d9e/authors.jar
2025-09-30 16:00:58,364 INFO mapreduce.ImportJobBase: Beginning import of authors
2025-09-30 16:00:58,365 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.job.tracker.address
2025-09-30 16:00:58,545 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2025-09-30 16:00:58,549 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 16:00:59,007 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2025-09-30 16:00:59,103 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2025-09-30 16:00:59,495 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_17592569
2025-09-30 16:01:02,350 INFO db.DBInputFormat: Using read committed transaction isolation
2025-09-30 16:01:02,351 INFO db.DataDriverDBInputFormat: BoundingValsQuery: SELECT MIN(id), MAX(id) FROM authors
2025-09-30 16:01:02,354 INFO db.IntegerSplitter: Split size: 1; Num splits: 4 from: 1 to: 8
2025-09-30 16:01:02,428 INFO mapreduce.JobSubmitter: number of splits:4
2025-09-30 16:01:02,587 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0006
2025-09-30 16:01:02,587 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-09-30 16:01:02,814 INFO conf.Configuration: resource-types.xml not found
2025-09-30 16:01:02,814 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'
2025-09-30 16:01:02,881 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0006
2025-09-30 16:01:02,920 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0006/
2025-09-30 16:01:02,921 INFO mapreduce.Job: Running job: job_1759256979608_0006
2025-09-30 16:01:11,086 INFO mapreduce.Job: Job job_1759256979608_0006 running in uber mode : false
2025-09-30 16:01:11,088 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 16:01:21,690 INFO mapreduce.Job: map 25% reduce 0%
2025-09-30 16:01:22,706 INFO mapreduce.Job: map 75% reduce 0%
2025-09-30 16:01:23,718 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 16:01:23,733 INFO mapreduce.Job: Job job_1759256979608_0006 completed successfully
2025-09-30 16:01:23,819 INFO mapreduce.Job: Counters: 34
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1127972
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=377
  HDFS: Number of bytes written=503
  HDFS: Number of read operations=24
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=8
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Killed map tasks=1
  Launched map tasks=4
  Other local map tasks=4
  Total time spent by all maps in occupied slots (ms)=30691
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=30691
  Total vcore-milliseconds taken by all map tasks=30691
  Total megabyte-milliseconds taken by all map tasks=31427584
Map-Reduce Framework
  Map input records=8
  Map output records=8
  Input split bytes=377
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=1383
  CPU time spent (ms)=5610
  Physical memory (bytes) snapshot=899338240
  Virtual memory (bytes) snapshot=11245617152
  Total committed heap usage (bytes)=723517440
  Peak Map Physical memory (bytes)=228921344
  Peak Map Virtual memory (bytes)=2814033920
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=503
2025-09-30 16:01:23,827 INFO mapreduce.ImportJobBase: Transferred 503 bytes in 24.8089 seconds (20.275 bytes/sec)
2025-09-30 16:01:23,832 INFO mapreduce.ImportJobBase: Retrieved 8 records.
[root@osboxes lib]# hdfs dfs -ls /mywarehouse/authors_compressed
Found 5 items
-rw-r--r-- 1 root supergroup 0 2025-09-30 16:01 /mywarehouse/authors_compressed/_SUCCESS
-rw-r--r-- 1 root supergroup 128 2025-09-30 16:01 /mywarehouse/authors_compressed/part-m-00000.gz
-rw-r--r-- 1 root supergroup 126 2025-09-30 16:01 /mywarehouse/authors_compressed/part-m-00001.gz
-rw-r--r-- 1 root supergroup 123 2025-09-30 16:01 /mywarehouse/authors_compressed/part-m-00002.gz
-rw-r--r-- 1 root supergroup 126 2025-09-30 16:01 /mywarehouse/authors_compressed/part-m-00003.gz
[root@osboxes lib]#

```

Рисунок 18 — Импорт со сжатием

5 ЭКСПОРТ В MARIADB

Подготовка таблицы к экспорту представлена на Рисунке 19.

```
[root@osboxes lib]# mysql -u student -p
mysql: Deprecated program name. It will be removed in a future release, use '/usr/bin/mariadb' instead
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 119
Server version: 11.6.1-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE TABLE IF NOT EXISTS labs.authors_export LIKE labs.authors;
Query OK, 0 rows affected (0.060 sec)

MariaDB [(none)]> TRUNCATE TABLE labs.authors_export;
Query OK, 0 rows affected (0.024 sec)

MariaDB [(none)]> Bye
[root@osboxes lib]# sqoop export --driver org.mariadb.jdbc.Driver --connect jdbc:mariadb://localhost/labs --username student --password student --table
```

Рисунок 19 — Подготовка таблицы

Процесс экспорта представлен на Рисунке 20.

```
[root@osboxes lib]# sqoop export --driver org.mariadb.jdbc.Driver --connect jdbc:mariadb://localhost/labs --username student --password student --table
authors_export --fields-terminated-by '\t' --export-dir /user/student/authors_Dorthy
2025-09-30 16:07:24.620 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2025-09-30 16:07:24.656 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2025-09-30 16:07:24.721 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriate connection manager is not being set (via
fy explicitly which connection manager should be used next time.
2025-09-30 16:07:24.737 INFO manager.SqlManager: Using default fetchSize of 1000
2025-09-30 16:07:24.738 INFO tool.CodeGenTool: Beginning code generation
2025-09-30 16:07:24.978 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors_export AS t WHERE 1=0
2025-09-30 16:07:25.000 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors_export AS t WHERE 1=0
2025-09-30 16:07:25.022 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/hadoop/hadoop
Note: /tmp/sqoop-root/compile/1fb56f20f20ac7452bb0b6b820601f8d/authors_export.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2025-09-30 16:07:26.372 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/1fb56f20f20ac7452bb0b6b820601f8d/authors_export.jar
2025-09-30 16:07:26.388 INFO mapreduce.ExportJobBase: Beginning export of authors_export
2025-09-30 16:07:26.388 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2025-09-30 16:07:26.582 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2025-09-30 16:07:27.315 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors_export AS t WHERE 1=0
2025-09-30 16:07:27.326 INFO Configuration.deprecation: mapred.reduce.tasks.speculative.execution is deprecated. Instead, use mapreduce.reduce.speculative
2025-09-30 16:07:27.332 INFO Configuration.deprecation: mapred.map.tasks.speculative.execution is deprecated. Instead, use mapreduce.map.speculative
2025-09-30 16:07:27.332 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2025-09-30 16:07:27.459 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2025-09-30 16:07:27.878 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1759256979608_0007
2025-09-30 16:07:32.201 INFO input.FileInputFormat: Total input files to process : 1
2025-09-30 16:07:32.684 INFO mapreduce.JobSubmitter: number of splits:4
2025-09-30 16:07:32.719 INFO Configuration.deprecation: mapred.map.tasks.speculative.execution is deprecated. Instead, use mapreduce.map.speculative
2025-09-30 16:07:33.231 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0007
2025-09-30 16:07:33.433 INFO conf.Configuration: resource-types.xml not found
2025-09-30 16:07:33.433 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-09-30 16:07:33.499 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0007
2025-09-30 16:07:33.549 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0007/
2025-09-30 16:07:39.652 INFO mapreduce.Job: Job job_1759256979608_0007 running in uber mode : false
2025-09-30 16:07:39.653 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 16:07:45.980 INFO mapreduce.Job: map 25% reduce 0%
2025-09-30 16:07:46.996 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 16:07:47.088 INFO mapreduce.Job: Job job_1759256979608_0007 completed successfully
2025-09-30 16:07:47.100 INFO mapreduce.Job: Counters: 33
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1127172
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=860
  HDFS: Number of bytes written=0
  HDFS: Number of read operations=19
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=0
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=4
  Data-local map tasks=4
  Total time spent by all maps in occupied slots (ms)=16940
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=16940
  Total vcore-milliseconds taken by all map tasks=16940
  Total megabyte-milliseconds taken by all map tasks=17346560
Map-Reduce Framework
  Map input records=1
  Map output records=1
  Input split bytes=651
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=853
  CPU time spent (ms)=3430
  Physical memory (bytes) snapshot=828084224
  Virtual memory (bytes) snapshot=11243864064
  Total committed heap usage (bytes)=17750272
  Peak Map Physical memory (bytes)=211218432
  Peak Map Virtual memory (bytes)=2811498496
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=0
2025-09-30 16:07:47.096 INFO mapreduce.ExportJobBase: Transferred 860 bytes in 19.7492 seconds (43.546 bytes/sec)
2025-09-30 16:07:47.100 INFO mapreduce.ExportJobBase: Exported 1 records.
```

Рисунок 20 — Экспорт таблицы

Результат экспорта представлен на Рисунке 21.

```
[root@osboxes lib]# mysql -u student -p -e "SELECT * FROM labs.authors_export;"
mysql: Deprecatcd program name. It will be removed in a future release, use '/usr/bin/mariadb' instead
Enter password:
+-----+-----+-----+-----+-----+-----+
| id | first_name | last_name | email | birthdate | added |
+-----+-----+-----+-----+-----+-----+
| 1 | Dorthy | Hicks | dorthy.hicks@example.com | 1990-02-14 | 2023-05-10 09:15:00 |
+-----+-----+-----+-----+-----+-----+
[root@osboxes lib]#
```

Рисунок 21 — Проверка экспорта

Таким образом, таблица экспортирована корректно.

6 САМОСТОЯТЕЛЬНАЯ РАБОТА

Создание директории mylabs представлено на Рисунке 22.

```
[root@osboxes lib]# su - hadoop
Last login: Tue Sep 30 14:29:13 EDT 2025 on pts/0
[hadoop@osboxes ~]$ hdfs dfs -mkdir /tmp/mylabs
[hadoop@osboxes ~]$ |
```

Рисунок 22 — Создание директории mylabs

Импорт столбцов id, title, published_at из таблицы posts представлен на Рисунке 23.

```
2025-09-30 16:23:08,176 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-09-30 16:23:08,391 INFO conf.Configuration: resource-types.xml not found
2025-09-30 16:23:08,391 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-09-30 16:23:08,453 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0009
2025-09-30 16:23:08,493 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0009/
2025-09-30 16:23:08,493 INFO mapreduce.Job: Running job: job_1759256979608_0009
2025-09-30 16:23:14,596 INFO mapreduce.Job: Job job_1759256979608_0009 running in uber mode : false
2025-09-30 16:23:14,597 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 16:23:20,769 INFO mapreduce.Job: map 50% reduce 0%
2025-09-30 16:23:22,790 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 16:23:23,835 INFO mapreduce.Job: Job job_1759256979608_0009 completed successfully
2025-09-30 16:23:23,950 INFO mapreduce.Job: Counters: 33
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1128200
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=377
  HDFS: Number of bytes written=266
  HDFS: Number of read operations=24
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=8
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=4
  Other local map tasks=4
  Total time spent by all maps in occupied slots (ms)=14000
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=14000
  Total vcore-milliseconds taken by all map tasks=14000
  Total megabyte-milliseconds taken by all map tasks=14336000
Map-Reduce Framework
  Map input records=8
  Map output records=8
  Input split bytes=377
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=316
  CPU time spent (ms)=3850
  Physical memory (bytes) snapshot=900452352
  Virtual memory (bytes) snapshot=11250077696
  Total committed heap usage (bytes)=779091968
  Peak Map Physical memory (bytes)=228773888
  Peak Map Virtual memory (bytes)=2813440000
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=266
2025-09-30 16:23:23,957 INFO mapreduce.ImportJobBase: Transferred 266 bytes in 17.966 seconds (14.8058 bytes/sec)
2025-09-30 16:23:23,961 INFO mapreduce.ImportJobBase: Retrieved 8 records.
[hadoop@osboxes ~]$ hdfs dfs -cat /tmp/mylabs/posts_info/part-m-00000 | head
1 Hello Sqoop 2023-06-01 09:00:00.0
2 HDFS Basics 2023-06-02 10:00:00.0
[hadoop@osboxes ~]$ |
```

Рисунок 23 — Импорт столбцов из таблицы posts

Команда, использованная для импорта представлена в Листинге 1.

Листинг 1 — Скрипт импорта столбцов из posts

```
sqoop import --driver org.mariadb.jdbc.Driver \
--connect jdbc:mariadb://localhost/labs \
--username student --password student \
--table posts \
```

Продолжение Листинга 1

```
--fields-terminated-by '\t' \  
--columns "id, title, published_at" \  
--target-dir /tmp/mylabs/posts_info
```

Импорт тех же данных в формате Parquet со сжатием Snappy представлен на Рисунке 24.

```
2025-09-30 16:26:07,807 INFO mapreduce.Job: map 0% reduce 0%  
2025-09-30 16:26:15,975 INFO mapreduce.Job: map 25% reduce 0%  
2025-09-30 16:26:17,004 INFO mapreduce.Job: map 50% reduce 0%  
2025-09-30 16:26:18,025 INFO mapreduce.Job: map 75% reduce 0%  
2025-09-30 16:26:19,034 INFO mapreduce.Job: map 100% reduce 0%  
2025-09-30 16:26:20,053 INFO mapreduce.Job: Job job_1759256979608_0010 completed successfully  
2025-09-30 16:26:20,138 INFO mapreduce.Job: Counters: 33  
File System Counters  
  FILE: Number of bytes read=0  
  FILE: Number of bytes written=1132468  
  FILE: Number of read operations=0  
  FILE: Number of large read operations=0  
  FILE: Number of write operations=0  
  HDFS: Number of bytes read=24093  
  HDFS: Number of bytes written=8970  
  HDFS: Number of read operations=200  
  HDFS: Number of large read operations=0  
  HDFS: Number of write operations=40  
  HDFS: Number of bytes read erasure-coded=0  
Job Counters  
  Launched map tasks=4  
  Other local map tasks=4  
  Total time spent by all maps in occupied slots (ms)=22700  
  Total time spent by all reduces in occupied slots (ms)=0  
  Total time spent by all map tasks (ms)=22700  
  Total vcore-milliseconds taken by all map tasks=22700  
  Total megabyte-milliseconds taken by all map tasks=23244800  
Map-Reduce Framework  
  Map input records=8  
  Map output records=8  
  Input split bytes=377  
  Spilled Records=0  
  Failed Shuffles=0  
  Merged Map outputs=0  
  GC time elapsed (ms)=1094  
  CPU time spent (ms)=8630  
  Physical memory (bytes) snapshot=1099444224  
  Virtual memory (bytes) snapshot=1131294104  
  Total committed heap usage (bytes)=1028128768  
  Peak Map Physical memory (bytes)=292540416  
  Peak Map Virtual memory (bytes)=2835378176  
File Input Format Counters  
  Bytes Read=0  
File Output Format Counters  
  Bytes Written=0  
2025-09-30 16:26:20,145 INFO mapreduce.ImportJobBase: Transferred 8.7598 KB in 22.0663 seconds (406.5027 bytes/sec)  
2025-09-30 16:26:20,148 INFO mapreduce.ImportJobBase: Retrieved 8 records.  
[hadoop@osboxes ~]$ hdfs dfs -ls /tmp/mylabs/posts_compressed/  
Found 6 items  
drwxr-xr-x - hadoop supergroup 0 2025-09-30 16:25 /tmp/mylabs/posts_compressed/.metadata  
drwxr-xr-x - hadoop supergroup 0 2025-09-30 16:26 /tmp/mylabs/posts_compressed/.signals  
-rw-r--r-- 1 hadoop supergroup 887 2025-09-30 16:26 /tmp/mylabs/posts_compressed/3313f349-9d6e-430e-acac-8717c0c7832e.parquet  
-rw-r--r-- 1 hadoop supergroup 896 2025-09-30 16:26 /tmp/mylabs/posts_compressed/843d227a-714d-4bdd-b392-de36a626d34d.parquet  
-rw-r--r-- 1 hadoop supergroup 911 2025-09-30 16:26 /tmp/mylabs/posts_compressed/f0d70225-eb44-4296-8bcb-b56a0e429448.parquet  
-rw-r--r-- 1 hadoop supergroup 896 2025-09-30 16:26 /tmp/mylabs/posts_compressed/fa077eab-faed-4cd2-baff-4a178e8afb65.parquet  
[hadoop@osboxes ~]$
```

Рисунок 24 — Импорт posts в parquet

Команда, использованная для импорта представлена в Листинге 2.

Листинг 2 — Скрипт импорта posts в parquet со сжатием Snappy

```
sqoop import --driver org.mariadb.jdbc.Driver \  
  --connect jdbc:mariadb://localhost/labs \  
  --username student --password student \  
  --table posts \  
  --fields-terminated-by '\t' \  
  --columns "id, title, published_at" \  
  --as-parquetfile \  
  --compression-codec org.apache.hadoop.io.compress.SnappyCodec \  
  --target-dir /tmp/mylabs/posts_compressed
```

Импорт столбцов id, first_name, last_name, birthdate из таблицы authors представлен на Рисунке 25.

```

2025-09-30 16:28:47.503 INFO manager.SqlManager: Using default fetchSize of 1000
2025-09-30 16:28:47.503 INFO tool.CodeGenTool: Beginning code generation
2025-09-30 16:28:47.730 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM authors AS t WHERE 1=0
2025-09-30 16:28:47.760 INFO org.apache.hadoop.mapreduce.Configuration.deprecation: HADOOP_MAPRED_HOME is /home/hadoop/hadoop
Note: /tmp/sqoop-hadoop/compile/cf90cb0eb4da712e4d9fafc9104666d9/authors.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2025-09-30 16:28:49.124 INFO org.apache.hadoop.mapreduce.Configuration.deprecation: Writing jar file: /tmp/sqoop-hadoop/compile/cf90cb0eb4da712e4d9fafc9104666d9/authors.jar
2025-09-30 16:28:49.813 INFO tool.ImportTool: Destination directory /user/student/authors is not present, hence not deleting.
2025-09-30 16:28:49.822 INFO mapreduce.ImportJobBase: Beginning import of authors
2025-09-30 16:28:49.833 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2025-09-30 16:28:49.828 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2025-09-30 16:28:49.840 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2025-09-30 16:28:49.951 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2025-09-30 16:28:50.365 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1759256979608_0011
2025-09-30 16:28:53.564 INFO db.DBInputFormat: Using read committed transaction isolation
2025-09-30 16:28:53.565 INFO db.DataDrivenDBInputFormat: BoundingValuesQuery: SELECT MIN(id), MAX(id) FROM authors
2025-09-30 16:28:53.569 INFO db.IntegerSplitter: Split size: 1; Num splits: 4; From: 1 to: 8
2025-09-30 16:28:54.049 INFO mapreduce.JobSubmitter: number of splits:4
2025-09-30 16:28:54.998 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0011
2025-09-30 16:28:54.988 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-09-30 16:28:54.839 INFO conf.Configuration: resource-types.xml not found
2025-09-30 16:28:54.839 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'
2025-09-30 16:28:54.912 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0011
2025-09-30 16:28:54.951 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0011/
2025-09-30 16:28:54.951 INFO mapreduce.Job: Running job: job_1759256979608_0011
2025-09-30 16:29:02.144 INFO mapreduce.Job: Job job_1759256979608_0011 running in uber mode : false
2025-09-30 16:29:02.145 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 16:29:02.145 INFO mapreduce.Job: map 25% reduce 0%
2025-09-30 16:29:02.145 INFO mapreduce.Job: map 75% reduce 0%
2025-09-30 16:29:11.453 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 16:29:11.517 INFO mapreduce.Job: Job job_1759256979608_0011 completed successfully
2025-09-30 16:29:11.515 INFO mapreduce.Job: Counters: 33
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1128268
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=377
  HDFS: Number of bytes written=200
  HDFS: Number of read operations=24
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=8
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=4
  Other local map tasks=4
  Total time spent by all maps in occupied slots (ms)=16352
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=16352
  Total vcore-millisecsk taken by all map tasks=16352
  Total megabyte-millisecsk taken by all map tasks=16744448
Map-Reduce Framework
  Map input records=8
  Map output records=8
  Input split bytes=377
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=430
  CPU time spent (ms)=4080
  Physical memory (bytes) snapshot=886325248
  Virtual memory (bytes) snapshot=11251490816
  Total committed heap usage (bytes)=21381760
  Peak Map Physical memory (bytes)=229457920
  Peak Map Virtual memory (bytes)=2814586880
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=200
2025-09-30 16:29:11.625 INFO mapreduce.ImportJobBase: Transferred 200 bytes in 21.7666 seconds (9.1884 bytes/sec)
2025-09-30 16:29:11.630 INFO mapreduce.ImportJobBase: Retrieved 8 records.
[hadoop@osboxes ~]$ hdfs dfs -ls /user/student/authors
Found 9 items
-rw-r--r-- 1 hadoop student 0 2025-09-30 16:29 /user/student/authors/_SUCCESS
-rw-r--r-- 1 hadoop student 51 2025-09-30 16:29 /user/student/authors/part-m-00000
-rw-r--r-- 1 hadoop student 49 2025-09-30 16:29 /user/student/authors/part-m-00001
-rw-r--r-- 1 hadoop student 49 2025-09-30 16:29 /user/student/authors/part-m-00002
-rw-r--r-- 1 hadoop student 51 2025-09-30 16:29 /user/student/authors/part-m-00003
[hadoop@osboxes ~]$

```

Рисунок 25 — Импорт столбцов из authors

Результат импорта представлен на Рисунке 26.

```

[hadoop@osboxes ~]$ hdfs dfs -cat /user/student/authors/part-m-00000 | head
1 Dorthy Hicks 1990-02-14
2 Ivan Petrov 1987-11-02
[hadoop@osboxes ~]$ hdfs dfs -cat /user/student/authors/part-m-00001 | head
3 Anna Smirnova 1995-07-30
4 John Doe 1980-01-05
^[[A[hadoop@osboxes ~]$ hdfs dfs -cat /user/student/authors/part-m-00002 | head
5 Maria Garcia 1992-03-18
6 Wei Zhang 1993-09-09
[hadoop@osboxes ~]$ hdfs dfs -cat /user/student/authors/part-m-00003 | head
7 Fatima Ali 1988-12-21
8 Elena Ivanova 1996-06-11
[hadoop@osboxes ~]$

```

Рисунок 26 — Результат импорта

Команда, использованная для импорта представлена в Листинге 3.

Листинг 3 — Скрипт импорта authors

```

sqoop import --driver org.mariadb.jdbc.Driver
--connect jdbc:mariadb://localhost/labs \
--username student --password student \
--table authors --fields-terminated-by '\t' \
--columns "id, first_name, last_name, birthdate" \
--target-dir /user/student/authors \
--delete-target-dir

```

Импорт не нулевых постов в parquet со сжатием Snappy представлен на

Рисунок 27.

```
[hadoop@osboxes ~]$ sqoop import --driver org.mariadb.jdbc.Driver --connect jdbc:mariadb://localhost/labs --username student --password student --target-dir /tmp/mylabs/posts_NotN
2025-09-30 16:32:46,615 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2025-09-30 16:32:46,653 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2025-09-30 16:32:46,722 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriate connection manager should be used next time.
2025-09-30 16:32:46,736 INFO manager.SqlManager: Using default fetchSize of 1000
2025-09-30 16:32:46,736 INFO tool.CodeGenTool: Beginning code generation
2025-09-30 16:32:46,736 INFO tool.CodeGenTool: Will generate java class as codegen_posts
2025-09-30 16:32:46,963 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM posts AS t WHERE 1=0
2025-09-30 16:32:46,997 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/hadoop/hadoop
Note: /tmp/sqoop-hadoop/compile/f62c52bbe0cb7c80ad674a14c45d710d/codegen_posts.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2025-09-30 16:32:48,169 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-hadoop/compile/f62c52bbe0cb7c80ad674a14c45d710d/codegen_posts.jar
2025-09-30 16:32:48,191 INFO mapreduce.ImportJobBase: Beginning import of posts
2025-09-30 16:32:48,192 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2025-09-30 16:32:48,402 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2025-09-30 16:32:48,934 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM posts AS t WHERE 1=0
2025-09-30 16:32:48,935 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM posts AS t WHERE 1=0
2025-09-30 16:32:49,124 WARN spi.Registration: Not loading URI patterns in org.kitesdk.data.spi.hive.Loader
2025-09-30 16:32:49,636 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2025-09-30 16:32:49,720 INFO client.DefaultHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2025-09-30 16:32:50,041 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging-hadoop-20250930163250041
2025-09-30 16:32:52,650 INFO db.DBInputFormat: Using read committed transaction isolation
2025-09-30 16:32:52,650 INFO db.DataDriverDBInputFormat: BoundingValsQuery: SELECT MIN(id), MAX(id) FROM posts WHERE (title is not null)
2025-09-30 16:32:52,655 INFO db.IntegerSplitter: Split size: 1; Num splits: 4 from: 1 to: 8
2025-09-30 16:32:52,713 INFO mapreduce.JobSubmitter: number of splits:4
2025-09-30 16:32:52,852 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1759256979608_0012
2025-09-30 16:32:52,852 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-09-30 16:32:53,076 INFO conf.Configuration: resource-types.xml not found
2025-09-30 16:32:53,076 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-09-30 16:32:53,136 INFO impl.YarnClientImpl: Submitted application application_1759256979608_0012
2025-09-30 16:32:53,172 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1759256979608_0012/
2025-09-30 16:32:53,172 INFO mapreduce.Job: Running job: job_1759256979608_0012
2025-09-30 16:33:00,282 INFO mapreduce.Job: Job job_1759256979608_0012 running in uber mode : false
2025-09-30 16:33:00,283 INFO mapreduce.Job: map 0% reduce 0%
2025-09-30 16:33:06,461 INFO mapreduce.Job: map 25% reduce 0%
2025-09-30 16:33:08,497 INFO mapreduce.Job: map 50% reduce 0%
2025-09-30 16:33:09,508 INFO mapreduce.Job: map 100% reduce 0%
2025-09-30 16:33:10,533 INFO mapreduce.Job: Job job_1759256979608_0012 completed successfully
2025-09-30 16:33:10,616 INFO mapreduce.Job: Counters: 33

File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=1133044
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=23597
  HDFS: Number of bytes written=9200
  HDFS: Number of read operations=200
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=40
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=4
  Other local map tasks=4
  Total time spent by all maps in occupied slots (ms)=20614
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=20614
  Total vcore-milliseconds taken by all map tasks=20614
  Total megabyte-milliseconds taken by all map tasks=21108736

Map-Reduce Framework
  Map input records=6
  Map output records=6
  Input split bytes=377
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=904
  CPU time spent (ms)=8260
  Physical memory (bytes) snapshot=1147498496
  Virtual memory (bytes) snapshot=11316789248
  Total committed heap usage (bytes)=1031274496
  Peak Map Physical memory (bytes)=290369536
  Peak Map Virtual memory (bytes)=2838859776

File Input Format Counters
  Bytes Read=0
  Bytes Written=0
2025-09-30 16:33:10,626 INFO mapreduce.ImportJobBase: Transferred 8.9844 KB in 20.9753 seconds (438.6101 bytes/sec)
2025-09-30 16:33:10,630 INFO mapreduce.ImportJobBase: Retrieved 6 records.
[hadoop@osboxes ~]$ hdfs dfs -cat /tmp/mylabs/posts_NotN/part-m-00000 | head
cat: /tmp/mylabs/posts_NotN/part-m-00000: No such file or directory
[hadoop@osboxes ~]$ hdfs dfs -ls /tmp/mylabs/posts_NotN/
Found 6 items
drwxr-xr-x - hadoop supergroup 0 2025-09-30 16:32 /tmp/mylabs/posts_NotN/.metadata
drwxr-xr-x - hadoop supergroup 0 2025-09-30 16:33 /tmp/mylabs/posts_NotN/.signals
-rw-r--r-- 1 hadoop supergroup 978 2025-09-30 16:33 /tmp/mylabs/posts_NotN/3082ff4f-d9b3-482d-99be-5e6616e01203.parquet
-rw-r--r-- 1 hadoop supergroup 938 2025-09-30 16:33 /tmp/mylabs/posts_NotN/425e8403-1f70-4882-865a-f00c88071421.parquet
-rw-r--r-- 1 hadoop supergroup 1015 2025-09-30 16:33 /tmp/mylabs/posts_NotN/464da249-f9bf-4d6e-9b41-a4ddbca4a7e.parquet
-rw-r--r-- 1 hadoop supergroup 953 2025-09-30 16:33 /tmp/mylabs/posts_NotN/baa761a9-08c4-4994-b80f-44723bc5bfea.parquet
[hadoop@osboxes ~]$
```

Рисунок 27 — Импорт не нулевых постов

Команда, использованная для импорта представлена в Листинге 4.

Листинг 4 — Скрипт импорта ненулевых постов

```
sqoop import --driver org.mariadb.jdbc.Driver
--connect jdbc:mariadb://localhost/labs \
--username student \
--password student \
```

Продолжение Листинга 4

```
--table posts \  
--where "title is not NULL" \  
--fields-terminated-by '\t' \  
--columns "id, title, content" \  
--as-parquetfile \  
--compression-codec org.apache.hadoop.io.compress.SnappyCodec \  
--target-dir /tmp/mylabs/posts NotN
```

Таким образом, самостоятельная работа выполнена.

ВЫВОД

В ходе выполнения практической работы была успешно реализована интеграция между реляционной базой данных MariaDB и распределенной файловой системой HDFS с использованием инструмента Apache Sqoop.

Были созданы и заполнены тестовыми данными таблицы `authors` и `posts` в базе данных `labs`.

Освоены базовые команды Sqoop для просмотра метаданных, таких как список баз данных и таблиц. Выполнены различные сценарии импорта данных из MariaDB в HDFS, включая импорт целых таблиц, выборку определенных столбцов, фильтрацию строк с помощью условия `WHERE`, а также сохранение данных в эффективном формате Parquet и с применением сжатия.

Также была продемонстрирована обратная операция — экспорт данных из HDFS обратно в таблицу MariaDB.

Несмотря на возникшие технические сложности на этапе импорта, связанные с совместимостью компонентов, все поставленные задачи, включая задания для самостоятельного выполнения, были в конечном итоге решены, что позволило получить практические навыки полного цикла работы с данными в экосистеме Hadoop.